Latent Feature Extraction for Musical Genres from Raw Audio

UE24CS352A Machine Learning, Mini-Project Write-up | Team: Amogh Sharma (PES1UG24CS053), Amartya Sinha (PES1UG24CS051)

1. Problem statement

Learn a compact encoding of musical genre directly from the raw audio waveform, with no hand-engineered features such as MFCCs or spectrograms. We follow Sawhney, Vasavada and Wang (Stanford CS229, 2018) [1], whose Deep Softmax Autoencoder (DSAE) compresses a short clip of audio into 64 numbers and must, from those 64 numbers alone, both reconstruct the clip and predict its genre. Our work has three parts: re-implement the paper's three models, examine how sound its evaluation is, and extend it with one training improvement and two new uses of the learned encodings.

2. Dataset

GTZAN Genre Collection [2], restricted as in the paper to four genres (classical, jazz, metal, pop): 400 songs of 30 s, mono, 22,050 Hz. Each song is cut into clips of 20,000 samples (0.9 s) and every clip is average-pooled with a window of 40, giving a 500-dimensional input; 20 clips per song give 8,000 examples. Inputs are scaled by one mean and one standard deviation from the training set. Nothing else is computed from the audio.

We evaluate under two protocols. The **random clip split** of the paper shuffles the clips into 6,000 / 1,000 / 1,000 for train, dev and test, so clips of one song fall on both sides. Our **song-level split** assigns whole songs (300 / 50 / 50), so test songs are never heard in training. GTZAN is known to contain repeated recordings [3]; we detect 30 duplicated pairs in these genres from their loudness envelopes and keep both copies in training. Before this correction, 5 of 50 test songs had a copy in the training set.

3. Approach

Model (from the paper)	Architecture, tanh units	Objective
Two Layer Neural Network	500 → 128 → 4	cross-entropy
Vanilla Autoencoder	500 → 256 → 192 → 128 → **64** → 128 → 192 → 256 → 500	reconstruction error (MSE)
Deep Softmax Autoencoder	the autoencoder, plus a classifier 64 → 32 → 16 → 4 on the encoding	0.9 x MSE + 0.1 x cross-entropy

Paper settings are kept: Adam, learning rate 1e-4, batch size 512, dropout 0.1, 500 epochs. Our additions:

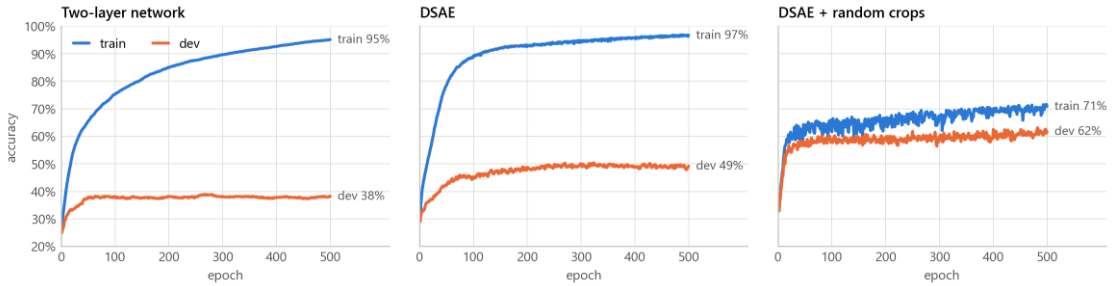
- **Random-crop training.** Instead of 20 fixed clips per song, every epoch draws 20 new windows at random positions from each training song, so no clip is seen twice. Learning rate 1e-3, selected on the dev set.
- **Diagnostic baselines.** A linear probe (logistic regression) on every representation, including a single hand-made number per clip, its loudness.
- **Novel feature 1, Genre morph.** Straight-line interpolation between the encodings of two clips; each intermediate point is classified and decoded. We measure how steadily the prediction changes and which genres are crossed. The paper names this as future work.
- **Novel feature 2, Similar-song search.** A song's fingerprint is the centred, normalised mean of its clip encodings; a library of 300 songs is ranked by cosine similarity. This is the external evaluation of the encodings that the paper calls for.

4. Implementation

PyTorch on CPU; each model trains in under five minutes. The `src` package separates configuration, data (loading, pooling, duplicate detection, splits, crop sampler), models, training, inference, evaluation, plotting, the two new features (`morph.py`, `retrieval.py`) and two experiment scripts (a sweep of the loss weight and a crops x learning-rate ablation). One command, `python -m src.evaluate`, regenerates every table and figure from the saved checkpoints. All randomness is seeded. 20 unit tests cover the pipeline, and a four-tab Streamlit app (`app.py`) gives the live demo: prediction with playback of what the network receives, Genre morph, Similar songs, and the results.

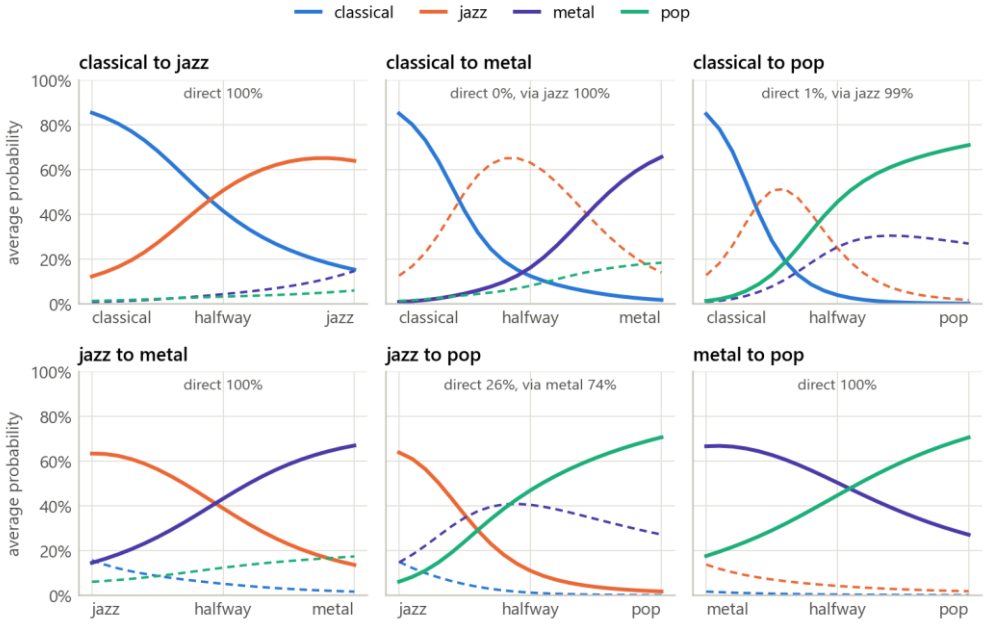
5. Results

Test accuracy on 0.9 s clips (chance 25%)	Paper reports	Ours, paper's split	Ours, unseen songs	Unseen songs + random crops
Two Layer Neural Network	36.4%	47.0%	40.5%	**66.8%**
Deep Softmax Autoencoder	65.3%	56.3%	52.2%	**66.8%**



Accuracy during training on the song-level split. Fixed clips are memorised (left, middle); random crops are not (right).

- **Memorisation.** With the paper's training the DSAE reaches 96.7% on training clips but 49.2% on dev. A linear probe on loudness alone scores 62.0%, above the paper-protocol DSAE; pooling to 551 values per second leaves only content below 276 Hz.
- **Random crops.** Accuracy on unseen songs rises from 52.2% to 66.8%, above the 65.3% the paper reports on its easier split, and the encoding now beats loudness under a linear probe (66.2%).
- **Genre morph.** The latent space orders the genres as a chain, classical, jazz, metal, pop, which is their order of loudness. 92% of latent walks change the prediction steadily, against 70% when the two audio inputs are cross-faded.
- **Similar-song search.** For 100 unseen query songs the nearest library song has the query's genre 68% of the time (loudness alone: 58%; vanilla autoencoder: 41%; chance: 25%).



Genre morph: average genre probabilities on the walk between the encodings of two genres. Solid lines are the end genres. Walks between distant genres pass through the genres that lie between them in loudness.

6. Conclusions

- The paper's ranking of models is reproduced, but its evaluation is optimistic: holding out whole songs and removing duplicated recordings lowers the DSAE from 56.3% to 52.2%.
- The dominant weakness is memorisation of a fixed set of clips. Random-crop training removes it without changing the architecture, the loss or the amount of computation.
- With random crops the plain two-layer network (66.8%) matches the DSAE (66.8%). The value of the DSAE is therefore its encoding with a decoder, not classification accuracy; both of our new features rely on it.
- The encodings are smooth and usable for retrieval, but they are organised mainly by loudness rather than by musical style. This follows from the band-limited input, which is the main limit of the method. Other limits: a small test set (50 songs) and remaining artist overlap in GTZAN.

[1] A. Sawhney, V. Vasavada, W. Wang. Latent Feature Extraction for Musical Genres from Raw Audio. Stanford CS229, 2018. [2] G. Tzanetakis, P. Cook. Musical Genre Classification of Audio Signals. IEEE Trans. Speech and Audio Processing, 2002. [3] B. L. Sturm. The GTZAN dataset: its contents, its faults, their effects on evaluation, and its future use. arXiv:1306.1461, 2013.